

Chapter

Photosynthesis

1. The process of conversion of ADP to ATP during photosynthesis is called:

MAIN 2024

- (a) Photolysis
- (b) Phagocytosis
- (c) Photophosphorylation
- (d) Polymerisation

2. A biologist studies how ADP is converted to ATP during photosynthesis under different light conditions. The process of conversion of ADP to ATP during Photosynthesis : SQP 2021

- (a) Polymerisation.
- (b) Photophosphorylation.
- (c) Photorespiration.
- (d) Photolysis.

3. What role do carotenoids play in photosynthesis? COMP 2005

- (a) They produce oxygen
- (b) They absorb and transfer light energy to chlorophyll
- (c) They convert light energy into chemical energy
- (d) They break down chlorophyll

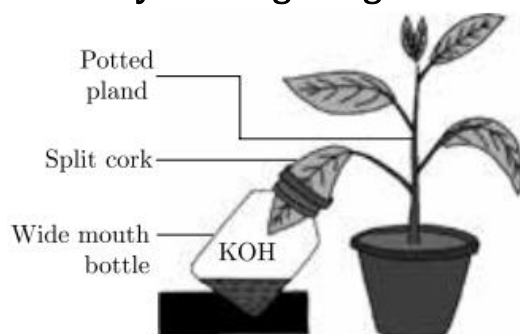
4. The site of dark reaction of photosynthesis. COMP 2021

5. The organisms which convert complex organic matter into simple ones.

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6. A plant that does not perform photosynthesis. COMP 2023
7. ____colour forms when iodine is added to starch. SQP 2021
8. Chlorophyll is present in the____of chloroplasts. COMP 2023
9. The immediate product of photosynthesis. MAIN 2003
10. The part of chloroplast where light reaction occurs. MAIN 2022
11. The ultimate electron donor in photosynthesis is water. COMP 2022
12. Land plants obtain their carbon dioxide from the atmosphere.
COMP 2023
13. Mention the significance of the following minerals: SQP 2025
- (a) Iodine in our food.
- (b) Magnesium for green plants.
14. Stomata SQP 2001
15. What is photophosphorylation? SQP 2021
16. What is photosynthesis? MAIN 2023
17. A destarched plant means[its leaves are without chlorophyll, plant is without chlorophyll, plant is without starch, aerial parts (mainly leaves) are without starch]. SQP 2022
18. ATP is needed for dark reaction.
19. Distinguish between Producers and Consumers. COMP 2022
20. Without changing the first term rearrange the remaining terms in logical sequence. Destarched plant, iodine added, washed in water, a leaf boiled in alcohol, placed in sunlight (testing the presence of starch). MAIN 2021

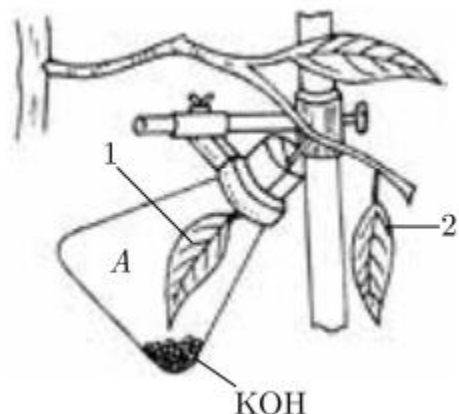
21. Study the diagram given below and fill in the blanks with suitable words:



In order to prove that carbon dioxide is necessary for (i) a potted plant is placed in dark for 48 hours to (ii) the leaves. A part of a leaf is inserted into a conical flask containing potassium hydroxide solution. This is to absorb (iii) from the flask. The plant is then placed in sunlight for a few hours. The experimental leaf is tested for starch. The part of the leaf that was inside the conical flask turns (iv) whereas the part of the leaf outside turns (v) in colour.

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22. The figure given below represents an experiment to demonstrate a particular aspect of photosynthesis. The alphabet 'A' represents a certain condition inside the flask. (i) What is the aim of the experiment ? SQP 2021



- (ii) Identify the special condition inside the flask.
- (iii) Name an alternative chemical that can be used instead of KOH.
- (iv) In what manner do the leaves 1 and 2 differ at the end of the starch test ?

Solution

1. (c) Photophosphorylation

The conversion of ADP to ATP during photosynthesis is called **Photophosphorylation**. (c) Photophosphorylation

2. (b) Photophosphorylation

It is the process of forming ATP from ADP using light energy during photosynthesis.

3. (b) They absorb and transfer light energy to chlorophyll.

Carotenoids absorb light energy and pass it to chlorophyll for photosynthesis.

4. Stroma

5. Decomposers

6. Mushroom/Fungus

7. Blue-black

8. Thylakoids

9. Glucose

10. Grana

11. True

12. True.

13. (a) Iodine – For making thyroxine.

(b) Magnesium – For making chlorophyll.

14. Stomata – Help in exchange of gases and transpiration.

15. Photophosphorylation – Conversion of ADP and inorganic phosphate into ATP in the presence of light.

16. Photosynthesis – The process by which green plants manufacture glucose from carbon dioxide and water in the presence of sunlight and chlorophyll.

17. A destarched plant means its aerial parts (mainly leaves) are without starch.

18. ATP is needed for dark reaction as it provides the energy required for the process.

19.

Producers	Consumers
(i) They are called autotrophs.	(i) They are called heterotrophs.
(ii) They can make their own food.	(ii) They cannot make their own food.
(iii) Their cells contain chlorophyll.	(iii) Chlorophyll is absent in their cells.

20. Destarched plant → placed in sunlight → a leaf boiled in alcohol → washed in water → iodine added.

21. (i) Photosynthesis

(ii) Destarch

(iii) Carbon dioxide

(iv) Brown

(v) Blue-black

22. (i) **Aim:** To show that carbon dioxide is necessary for photosynthesis.

(ii) **Condition inside flask (A):** KOH absorbs all the carbon dioxide, so the leaf inside does not get CO₂.

(iii) **Alternative chemical:** Sodium hydroxide (NaOH).

(iv) **Difference in leaves:**

- Leaf 1 – Does not turn blue-black (no starch formed).
- Leaf 2 – Turns blue-black (starch formed).